Data Driven

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Master's in Data Science, March 2025

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1.1 The Act of Reasoning

Reasoning is the mental process through which we derive conclusions from premises. In both everyday situations and scientific inquiry, we reason to argue, generalize, explain, or interpret.

Based on Mantere & Ketokivi (2013), scientific reasoning can be categorized into three core types, often illustrated using the metaphor of marbles in a box:

- **Inductive Reasoning (Stricter Sense):** Starts from observations and aims to generalize a rule.

  Observation: These marbles are red
  Explanation: These marbles are from this box
  $\Rightarrow$ **Rule:** All marbles in the box are red

  Inductive Reasoning generalizes from particular cases. It relies on patterns in empirical data to propose theoretical rules. However, inductive generalizations are never logically certain — this is known as the *problem of induction* (Popper, 1959).

  Inductive Research builds theory from contextual data, often through case studies.

- **Deductive Reasoning:** Starts from a general rule and uses it, along with an explanation, to derive an observation.
  Example:

  Rule: All marbles in this box are red
  Explanation: These marbles are from this box
  $\Rightarrow$ **Observation:** The marbles are red

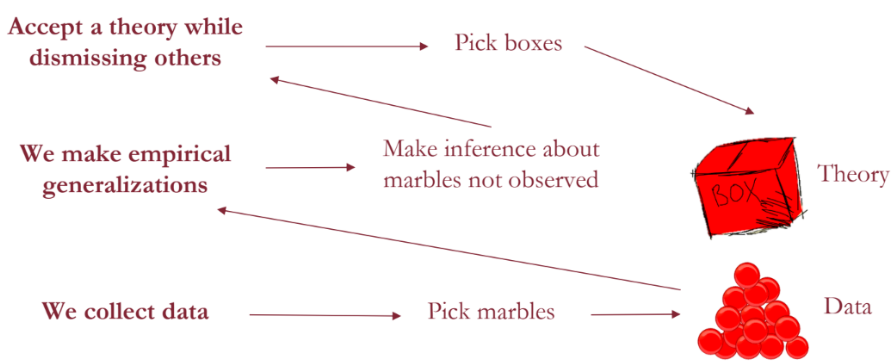
  Deductive Reasoning, on the other hand, starts from a theory or rule and applies it to specific instances. It ensures logical consistency; if the premises are true, the conclusion must be true.

  Deductive Research tests pre-existing hypotheses using statistical inference.

- <u>**Abductive Reasoning:**</u> Starts from an observation and a rule to suggest an explanation.

Observation: These marbles are red
Rule: All marbles in this box are red
$\Rightarrow$ **Explanation:** These marbles are from this box

These reasoning types help frame how we approach scientific analysis: gathering data (e.g., picking marbles), making generalizations (induction), and choosing the best theories (deduction and abduction).



1.2 What is Econometrics?

Econometrics applies statistical methods to economic data in order to empirically validate theoretical models.

> *Ëconometrics is the application of statistical methods to economic data in order to give empirical content to economic relationships.¨*
> (Pesaran, 1987)

1.2.1 Economic vs. Econometric Models

Economic Models describe relationships using mathematical equations based on assumptions (e.g., utility maximization).

Example: Becker's model of crime assumes rational individuals weigh the costs and benefits of criminal behavior:

- Benefits: earnings from illegal activity
- Costs: opportunity cost, probability of arrest and conviction, sentence length

The resulting behavioral function:

$$y = f(x_1, x_2, x_3, x_4, x_5, x_6, x_7)$$

where:

$$\begin{aligned}
y &= \text{Hours in criminal activity} \\
x_1 &= \text{Wage in criminal activity} \\
x_2 &= \text{Wage in legal work} \\
x_3 &= \text{Other income} \\
x_4 &= \text{Probability of getting caught} \\
x_5 &= \text{Probability of conviction} \\
x_6 &= \text{Expected sentence} \\
x_7 &= \text{Age}
\end{aligned}$$

Econometric Models translate this into an estimable form:

$$\text{crime} = \beta_0 + \beta_1\text{wage} + \beta_2\text{othinc} + \beta_3\text{freqarr} + \beta_4\text{freqconv} + \beta_5\text{avgsen} + \beta_6\text{age} + u$$

Note: The error term u captures unobserved variables like morality, family background, or unobservable wages.

1.2.2 Causality and *Ceteris Paribus*

Establishing causality is fundamental in policy and economic analysis.

Ceteris paribus means holding all else constant — assessing the effect of one variable while controlling for others.

Example: Does hospital care improve health?

Group	Sample Size	Mean Health Status	Standard Error
Hospital	7,774	2.79	0.014
No Hospital	90,049	2.07	0.003

Figure 1.1: Compare hospitalized vs. non-hospitalized individuals using 2005 NHIS data

- Hospitalized have worse average health: mean difference of 0.71.
- Problem: selection bias — sick people are more likely to go to the hospital.

Potential Outcomes Framework

Let $D_i = 1$ if individual i is hospitalized, 0 otherwise. Let Y_i be the observed health outcome. We define two potential outcomes:

$$Y_i = \begin{cases} Y_{1i} & \text{if } D_i = 1 \\ Y_{0i} & \text{if } D_i = 0 \end{cases} = Y_{0i} + (Y_{1i} - Y_{0i})D_i$$

Selection Bias and the Causal Effect

Goal: Estimate the **Average Treatment Effect on the Treated (ATT)**:

$$\mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1]$$

However, we face a fundamental problem: for each individual i, we only observe one of the two potential outcomes — either Y_{1i} (if treated) or Y_{0i} (if untreated), but never both.

Thus, when we compare average outcomes across treated and untreated individuals, the observed difference does not equal the ATT directly, because of possible systematic differences between the groups.

The decomposition is:

$$\underbrace{\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]}_{\text{Observed difference in average outcomes}} = \underbrace{\mathbb{E}[Y_{1i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]}_{\text{Selection Bias}}$$

- The **first term** is the true treatment effect on those who actually received treatment.
- The **second term** captures the difference in baseline outcomes between treated and untreated groups — this is the **selection bias**.

When Random Assignment Solves the Problem

Under a randomized experiment, treatment assignment D_i is statistically independent of the potential outcomes (Y_{0i}, Y_{1i}). This implies:

$$\mathbb{E}[Y_{0i} \mid D_i = 1] = \mathbb{E}[Y_{0i} \mid D_i = 0]$$

Thus, selection bias disappears and the observed difference in outcomes directly estimates the average treatment effect:

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \mathbb{E}[Y_{1i} - Y_{0i}]$$

Nonexperimental Data

In economics, most data are nonexperimental — not generated by controlled experiments, but observed from real-world behavior.

- Examples: government surveys, administrative records, financial databases
- Limitations: potential confounding, lack of randomization

Econometrics emerged to handle these limitations, focusing on causality from observational data.

Chapter 2

Lecture 3

2.1 Simple Linear Regression Model

The simple linear regression model relates a dependent variable y to a single independent variable x:

$$y_i = \beta_0 + \beta_1 x_i + u_i, \text{ where:}$$

- y_i: the dependent (explained) variable
- x_i: the independent (explanatory) variable
- u_i: the error term (disturbance), capturing unobserved factors
- β_0: intercept of the regression line
- β_1: slope of the regression line (effect of x on y)

Key assumptions:

- $\mathbb{E}[u_i] = 0$: the error has zero mean
- $\mathbb{E}[u_i|x_i] = 0$: the error is mean independent of x_i

These imply that the conditional expectation of y given x is linear in x.:

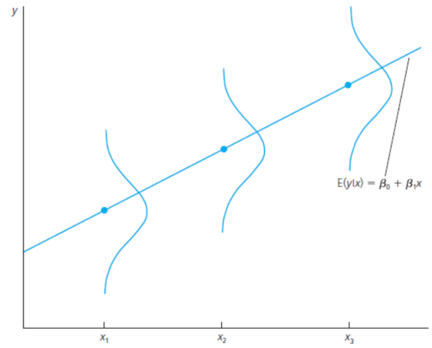
$$\mathbb{E}[y_i|x_i] = \beta_0 + \beta_1 x_i$$

We can decompose the model into:
Systematic component: $\beta_0 + \beta_1 x_i$, **Unsystematic component:** u_i

2.2 Ordinary Least Squares (OLS) Estimation

We observe a random sample of size n:

$$\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$$



The goal of OLS is to estimate β_0 and β_1 by minimizing the sum of squared residuals:

$$\min_{\beta_0,\beta_1} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

More in general:

$$\sum_{i=1}^{n} \hat{u}_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

This method is optimal under the classical assumptions because it leads to unbiased and efficient estimators.

2.2.1 First-Order Conditions (Method of Moments Approach)

By setting the derivatives of the sum of squared residuals with respect to β_0 and β_1 to zero, we obtain the normal equations. Solving them gives:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\widehat{\text{Cov}}(x, y)}{\widehat{\text{Var}}(x)}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

These are the OLS estimators for the intercept and slope.

Interpretation

- $\hat{\beta}_1$: measures the average change in y associated with a one-unit increase in x
- $\hat{\beta}_0$: the predicted value of y when $x = 0$

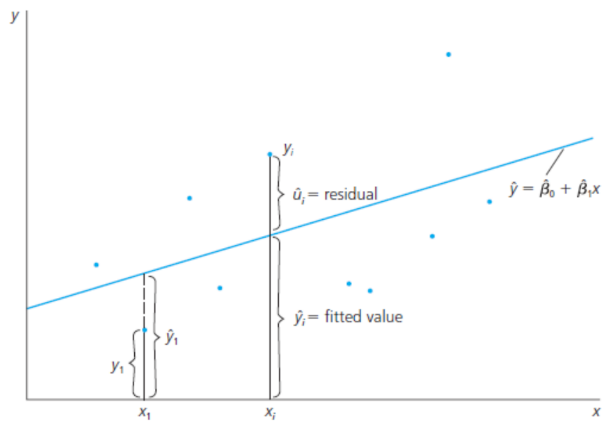
2.2.2 Properties of OLS Residuals (By Construction)

Let the fitted values be:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

and the residuals:

$$\hat{u}_i = y_i - \hat{y}_i$$



By construction, OLS residuals satisfy the following properties:

1. **The sum of residuals is zero:**

$$\sum_{i=1}^{n} \hat{u}_i = 0 \quad \Rightarrow \quad \frac{1}{n}\sum_{i=1}^{n} \hat{u}_i = 0$$

2. **The residuals are uncorrelated with the regressor:**

$$\sum_{i=1}^{n} x_i \hat{u}_i = 0 \quad \Rightarrow \quad \widehat{\text{Cov}}(x, \hat{u}) = 0$$

3. **The point $(\bar{x}, \bar{y})$ lies on the regression line:**

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y}$$

2.2.3 Goodness-of-Fit

We can decompose the total variation in y as:

$$y_i = \hat{y}_i + \hat{u}_i$$

Decomposition of Variance

- **Total Sum of Squares (SST):**

$$\text{SST} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

- **Explained Sum of Squares (SSE):**

$$\text{SSE} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

- **Residual Sum of Squares (SSR):**

$$\text{SSR} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \hat{u}_i^2$$

These satisfy the identity:

$$\text{SST} = \text{SSE} + \text{SSR}$$

2.2.4 Coefficient of Determination (R^2)

$$R^2 = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}}$$

Interpretation: R^2 measures the proportion of the total variation in y that is explained by the regression model.

Caution: A low R^2 does not necessarily imply a useless model. It could still yield unbiased and meaningful estimates of causal effects.

2.3 Assumptions for Unbiasedness of OLS (SLR)

The following assumptions are required to prove that OLS is unbiased:

- **SLR.1 — Linearity in Parameters:**

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

The relationship is linear in the parameters β_0 and β_1, not necessarily in x.

- **SLR.2 — Random Sampling:**

$$\{(x_i, y_i)\}_{i=1}^{n} \text{ is a random sample from the population}$$

Each observation follows the same model.

- **SLR.3 — Sample Variation in the Explanatory Variable:**

$$\text{Var}(x) > 0$$

The values of x are not all the same in the sample — there is variation to identify β_1.

- **SLR.4 — Zero Conditional Mean:**

$$\mathbb{E}[u_i \mid x_i] = 0$$

The error has an expected value of zero given any value of x_i. This is the most critical assumption for unbiasedness.

Unbiasedness of OLS Estimators

Under assumptions SLR.1 through SLR.4:

$$\mathbb{E}[\hat{\beta}_1] = \beta_1 \quad \text{and} \quad \mathbb{E}[\hat{\beta}_0] = \beta_0$$

That is, both OLS estimators are **unbiased** — on average, they hit the true population values.

Sketch of Proof:

To show that OLS provides unbiased estimates, we focus on the slope estimator $\hat{\beta}_1$.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\text{SST}_x}$$

Substitute $y_i = \beta_0 + \beta_1 x_i + u_i$ into the formula:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(\beta_0 + \beta_1 x_i + u_i)}{\text{SST}_x}$$

Distribute the term in the numerator:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})\beta_0 + \sum_{i=1}^n (x_i - \bar{x})\beta_1 x_i + \sum_{i=1}^n (x_i - \bar{x}) u_i}{\text{SST}_x}$$

Note that:

- $\sum_{i=1}^n (x_i - \bar{x}) = 0$, so the first term vanishes.
- $\sum_{i=1}^n (x_i - \bar{x}) x_i = \text{SST}_x$, so the second term becomes $\beta_1 \text{SST}_x$.

Therefore:

$$\hat{\beta}_1 = \frac{\beta_1 \text{SST}_x + \sum_{i=1}^n (x_i - \bar{x}) u_i}{\text{SST}_x} = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\text{SST}_x}$$

Let $d_i = x_i - \bar{x}$. Then:

$$\hat{\beta}_1 = \beta_1 + \frac{1}{\text{SST}_x} \sum_{i=1}^n d_i u_i$$

Conclusion: Under the zero conditional mean assumption ($\mathbb{E}[u_i \mid x_i] = 0$), it follows that:

$$\mathbb{E}[\hat{\beta}_1] = \beta_1$$

Thus, the OLS slope estimator is **unbiased**.

2.4 Variance of OLS Estimators

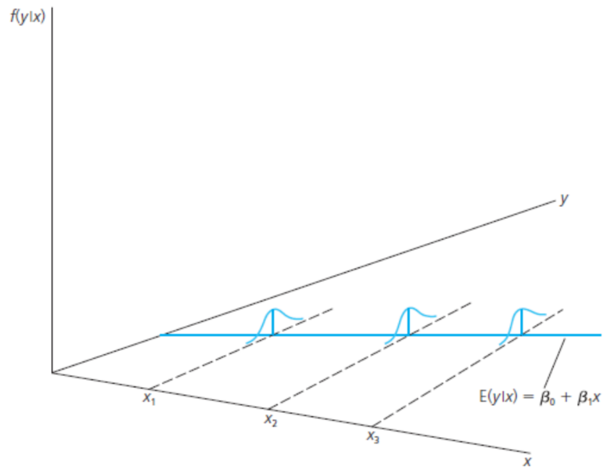
To quantify the precision of the OLS estimators, we need to compute their sampling variances.

SLR.5 — Homoskedasticity Assumption

The variance of the error term u_i, conditional on x_i, is constant:

$$\text{Var}(u_i \mid x_i) = \sigma^2 \quad \text{for all } i$$

This implies that the spread of the error term does not depend on the value of the explanatory variable x. When this assumption fails, we say that the errors are **heteroskedastic**.



Variance of the OLS Slope Estimator

Under assumptions SLR.1 through SLR.5:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\sigma^2}{\text{SST}_x}$$

Similarly, for the intercept:

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \right]$$

Interpretation:

- The greater the sample variation in x, the more precise (lower variance) is $\hat{\beta}_1$.
- If the sample size n increases, the variance of both estimators decreases.

Estimating the Error Variance

Since σ^2 is unknown, we estimate it using the residuals:

$$\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^{n} \hat{u}_i^2$$

Why divide by $n - 2$?

- We lose 2 degrees of freedom from estimating β_0 and β_1
- This adjustment ensures that $\hat{\sigma}^2$ is an unbiased estimator of σ^2

This estimator is unbiased for σ^2 under the classical assumptions.

Standard Error of the Regression (SER)

The square root of $\hat{\sigma}^2$ is known as the **SER**:

$$\text{SER} = \sqrt{\hat{\sigma}^2}$$

The SER is not the standard deviation of $\hat{\beta}_1$, but it is essential in computing the *standard errors* of the OLS coefficients.

2.5 Multiple Linear Regression (MLR) Model

The multiple linear regression model allows us to explain the dependent variable y using more than one explanatory variable. The population model is:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + u_i$$

where:

- y_i: dependent (explained) variable
- $x_{1i}, \ldots, x_{ki}$: independent (explanatory) variables
- β_0: intercept
- β_j: partial effect of x_j on y, for $j = 1, \ldots, k$
- u_i: error term, representing omitted factors

Key Assumptions:

- $\mathbb{E}[u_i] = 0$
- $\mathbb{E}[u_i \mid x_{1i}, \ldots, x_{ki}] = 0$
- $\mathbb{E}[y_i \mid x_{1i}, \ldots, x_{ki}] = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki}$

Objective: Estimate the unknown parameters $\beta_0, \ldots, \beta_k$ using sample data.

2.6 OLS Estimation in Multiple Regression

Suppose we observe a random sample of size n:

$$\{(y_i, x_{1i}, x_{2i}, \dots, x_{ki})\}_{i=1}^n$$

The OLS estimator minimizes the sum of squared residuals:

$$\min_{\beta_0,\beta_1,\dots,\beta_k} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^k \beta_j x_{ji} \right)^2$$

Solving this optimization problem yields:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

where $\mathbf{X}$ is the $n \times (k+1)$ matrix of regressors (including a column of 1s for the intercept), and $\mathbf{y}$ is the $n \times 1$ vector of outcomes.

Interpretation of Coefficients

Each $\hat{\beta}_j$ measures the partial effect of x_j on y, **holding all other regressors constant**. That is:

$$\hat{\beta}_j = \frac{\partial y}{\partial x_j}\bigg|_{\text{all other } x \text{ fixed}}$$

This is a key distinction from simple regression, where the estimated slope includes any confounding effects.

2.6.1 Partialling-Out Interpretation

The coefficient $\hat{\beta}_j$ in a multiple regression model can be understood as the effect of x_j on y, removing the variation in x_j that is explained by the other regressors.

Formally:

- Regress x_j on the other regressors $\{x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_k\}$, and collect the residuals.
- Regress y on the same set of regressors and collect the residuals.
- Regress the residuals of y on the residuals of x_j: the coefficient from this regression is exactly $\hat{\beta}_j$.

This is known as the **Frisch–Waugh–Lovell (FWL) Theorem**, and it shows that:

$$\hat{\beta}_j = \text{effect of the variation in } x_j \text{ that is orthogonal to other } x\text{'s on } y$$

Key implication: In multiple regression, each coefficient captures a *ceteris paribus* effect — holding other variables fixed.

2.6.2 Simple vs. Multiple Regression Estimates

In simple regression, the estimate $\hat{\beta}_1^{\text{simple}}$ captures:

$$\text{Total association between } x_1 \text{ and } y$$

In multiple regression, the estimate $\hat{\beta}_1^{\text{multiple}}$ captures:

$$\text{Partial effect of } x_1 \text{ on } y \text{ holding all other } x_j \text{ constant}$$

Example: Suppose income y depends on education x_1 and experience x_2. A simple regression of y on x_1 alone will capture both the effect of education and any indirect association due to experience (if education and experience are correlated).

In contrast, the multiple regression controls for x_2, isolating the true marginal impact of education on income.

Conclusion: Simple regression estimates may suffer from **omitted variable bias**, while multiple regression helps address this by controlling for confounders.

2.6.3 Assumptions for Unbiasedness of OLS in MLR

To ensure that OLS is unbiased in the multiple regression model, we require the following assumptions:

- **MLR.1 — Linearity in parameters:**

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + u_i$$

- **MLR.2 — Random sampling:**

$$\{(y_i, x_{1i}, \ldots, x_{ki})\}_{i=1}^n \text{ is a random sample}$$

- **MLR.3 — No perfect multicollinearity:**

$$\text{None of the regressors is a perfect linear combination of the others}$$

- **MLR.4 — Zero conditional mean:**

$$\mathbb{E}[u_i \mid x_{1i}, \ldots, x_{ki}] = 0$$

  This implies that the error term is uncorrelated with each explanatory variable.

Result: Under MLR.1–MLR.4, the OLS estimators are **unbiased**:

$$\mathbb{E}[\hat{\beta}_j] = \beta_j \quad \text{for all } j = 0, 1, \ldots, k$$

2.6.4 Omitted Variable Bias (OVB)

Omitted variable bias arises when a relevant explanatory variable is left out of the regression model and is correlated with both the dependent variable and an included regressor.

Formal Setting

Suppose the true model is:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 z_i + u_i$$

but we estimate:

$$y_i = \alpha_0 + \alpha_1 x_{1i} + e_i$$

Problem: If z_i is omitted and z_i is correlated with x_{1i}, then $\hat{\alpha}_1$ will be a biased estimator of β_1.

Bias Formula

If we regress y on x_1 only, and omit z, then the bias in $\hat{\alpha}_1$ is:

$$\text{Bias}(\hat{\alpha}_1) = \beta_2 \cdot \delta_1$$

where δ_1 is the slope from regressing z on x_1:

$$z_i = \delta_0 + \delta_1 x_{1i} + r_i$$

Interpretation:

- The bias is positive if $\beta_2 > 0$ and $\delta_1 > 0$ (i.e., omitted variable has a positive effect on y and is positively correlated with x_1)
- The direction and magnitude of the bias depend on both the omitted variable's true effect and its correlation with the included regressor

Conclusion: Leaving out relevant variables leads to biased and inconsistent estimates unless the omitted variable is uncorrelated with all included regressors.

2.6.5 Variance of OLS in Multiple Regression

Under assumptions MLR.1–MLR.5, we can derive the variance of each OLS slope estimator $\hat{\beta}_j$ for $j = 1, \ldots, k$.

General Variance Formula

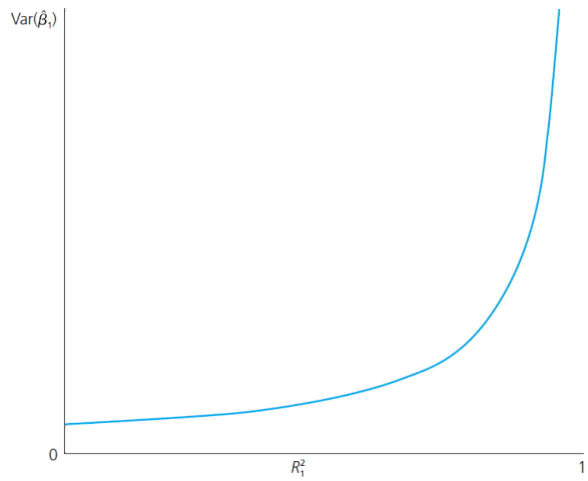
$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{SST_j \cdot (1 - R_j^2)}$$

where:

- $\sigma^2 = \text{Var}(u_i)$: error variance (assumed constant under homoskedasticity)
- $SST_j = \sum_{i=1}^{n} (x_{ji} - \bar{x}_j)^2$: sample variation in regressor x_j
- R_j^2: R-squared from an auxiliary regression of x_j on all other regressors

Interpretation

- If $R_j^2 \to 1$, then x_j is highly collinear with the other x's $\to \text{Var}(\hat{\beta}_j) \to \infty$
- If $R_j^2 = 0$, then x_j is orthogonal to the others $\to$ best-case precision



Multicollinearity and Variance Inflation Factor (VIF)

Multicollinearity arises when one regressor is highly linearly correlated with other regressors. This increases the variance of $\hat{\beta}_j$, making estimates less precise.

To quantify this, we define the Variance Inflation Factor (VIF) for each slope coefficient:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

Interpretation: VIF_j measures how much the variance of $\hat{\beta}_j$ is inflated due to multicollinearity.

Rules of thumb:

- $\text{VIF}_j = 1$: no multicollinearity
- $\text{VIF}_j > 5$: moderate multicollinearity
- $\text{VIF}_j > 10$: problematic multicollinearity (though threshold is arbitrary)

Unbiasedness vs. Variance in Multiple Regression

Suppose the population model is:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$$

We are interested in estimating β_1 using two approaches:

1. The **multiple regression estimator** $\hat{\beta}_1^{(\text{MR})}$, obtained from the full model with both x_1 and x_2
2. The **simple regression estimator** $\hat{\beta}_1^{(\text{SR})}$, obtained by omitting x_2 from the model

Bias–Variance Trade-Off

- $\hat{\beta}_1^{(\text{MR})}$ is **unbiased** under MLR.1–MLR.4
- $\hat{\beta}_1^{(\text{SR})}$ is **biased** if x_1 and x_2 are correlated, due to omitted variable bias
- However, $\text{Var}(\hat{\beta}_1^{(\text{SR})}) < \text{Var}(\hat{\beta}_1^{(\text{MR})})$ unless x_1 and x_2 are uncorrelated

Conclusion: The choice between estimators involves a trade-off:

$$\text{Lower bias} \quad \Leftrightarrow \quad \text{Higher variance}$$

$$\text{Lower variance} \quad \Leftrightarrow \quad \text{Potential bias}$$

This illustrates why we must carefully consider which variables to include: omitting relevant variables biases estimates, while including irrelevant ones increases variance but preserves unbiasedness.

Variance of OLS and Standard Error in MLR

Given the model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + u_i$$

An **unbiased estimator of the error variance** under MLR.1–MLR.5 is:

$$\hat{\sigma}^2 = \frac{1}{n-k-1} \sum_{i=1}^{n} \hat{u}_i^2$$

The **Standard Error of the Regression (SER)** is:

$$\text{SER} = \sqrt{\hat{\sigma}^2}$$

Effect of Adding Regressors

When a new explanatory variable is added to the model:

- The residual sum of squares (SSR) weakly decreases
- The degrees of freedom $(n-k-1)$ decreases
- Therefore, SER may increase or decrease depending on the relative change

Note: The usual estimator $\hat{\sigma}^2$ is **not valid under heteroskedasticity**. If the error variance is not constant:

- OLS remains unbiased
- But the estimated variances of $\hat{\beta}_j$ will be biased
- Solution: use robust standard errors

2.7 The Gauss-Markov Theorem (Paper at the end)

Statement: Under assumptions MLR.1–MLR.5, the OLS estimator $\hat{\boldsymbol{\beta}}$ is the **Best Linear Unbiased Estimator (BLUE)**.

MLR.5 — Homoskedasticity

$$\text{Var}(u_i \mid x_{1i}, \ldots, x_{ki}) = \sigma^2 \quad \text{for all } i$$

The error term has constant variance across all observations.

Implications of the Theorem

- **Best:** among all linear and unbiased estimators, OLS has the smallest variance
- **Linear:** OLS estimates are linear functions of the dependent variable y
- **Unbiased:** $\mathbb{E}[\hat{\beta}_j] = \beta_j$

Important: The Gauss-Markov Theorem does *not* require normality of errors. It only applies to linear estimators — other nonlinear estimators may be better, but not in the class of BLUE.

Violation: Heteroskedasticity

If MLR.5 fails (heteroskedasticity), OLS is still unbiased but no longer efficient (i.e., it is no longer the lowest variance estimator). This motivates the use of robust standard errors or alternative estimators.

Chapter 3

Lecture 4

3.1 Inference in the Multiple Regression Model

Assumption MLR.6 — Normality of Errors

In addition to assumptions MLR.1–MLR.5, we now introduce:

MLR.6: The population error term u is independent of the explanatory variables $x_1, \ldots, x_k$, and follows a normal distribution:

$$u \sim \mathcal{N}(0, \sigma^2)$$

Implications:

- This is a stronger assumption than MLR.4 and MLR.5.
- It implies that conditional on the regressors, the OLS estimators follow a normal distribution.

Together, MLR.1 through MLR.6 define the **Classical Linear Model (CLM)**. Under the CLM assumptions:

$$\hat{\beta}_j \sim \mathcal{N}(\beta_j, \mathrm{Var}(\hat{\beta}_j))$$

This enables us to perform hypothesis tests and construct confidence intervals for the β_j's.

3.1.1 t Distribution and Hypothesis Testing

Under the Classical Linear Model (MLR.1–MLR.6), each OLS coefficient $\hat{\beta}_j$ follows a normal distribution, and the standardized statistic:

$$t_j = \frac{\hat{\beta}_j - \beta_j^0}{\mathrm{SE}(\hat{\beta}_j)}$$

follows a Student's t-distribution with $n - k - 1$ degrees of freedom, where:

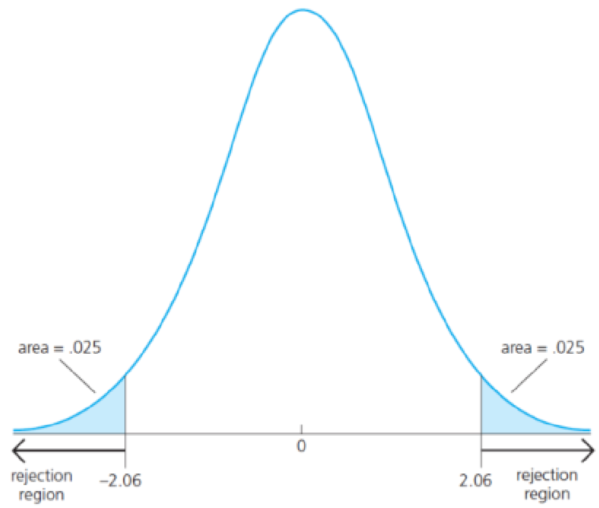


Figure 3.1: For a two tailed test, c is chosen to make the area in each tail of the t distribution equal 2.5%. In other words, c is the 97.5th percentile in the t distribution with n-k-1 degrees of freedom. When n-k-1=25, the 5% critical value for a two- sided test is c=2.060.

- β_j^0: hypothesized value of the parameter
- $\text{SE}(\hat{\beta}_j)$: standard error of $\hat{\beta}_j$

We use this distribution to test hypotheses on individual coefficients.

Two-Sided Hypothesis Testing

We want to test:

$$H_0 : \beta_j = \beta_j^0 \quad \text{vs.} \quad H_1 : \beta_j \neq \beta_j^0$$

The decision rule (at significance level α) is:

$$\text{Reject } H_0 \text{ if } |t_j| > t_{n-k-1,1-\alpha/2}$$

The critical value comes from the t-distribution with $n - k - 1$ degrees of freedom.

One-Sided Hypothesis Testing

$$H_0 : \beta_j \leq \beta_j^0 \quad \text{vs.} \quad H_1 : \beta_j > \beta_j^0$$

or

$$H_0 : \beta_j \geq \beta_j^0 \quad \text{vs.} \quad H_1 : \beta_j < \beta_j^0$$

We reject H_0 at level α if:

$$t_j > t_{n-k-1,1-\alpha} \quad \text{(right-tailed)}$$

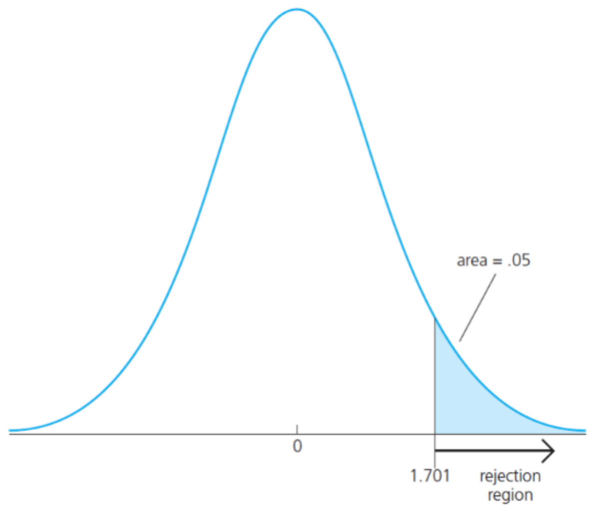


Figure 3.2: For a 5% significance level test and with n-k-1=28 degrees of freedom, the critical value is c=1.701.

$$t_j < -t_{n-k-1,1-\alpha} \quad \text{(left-tailed)}$$

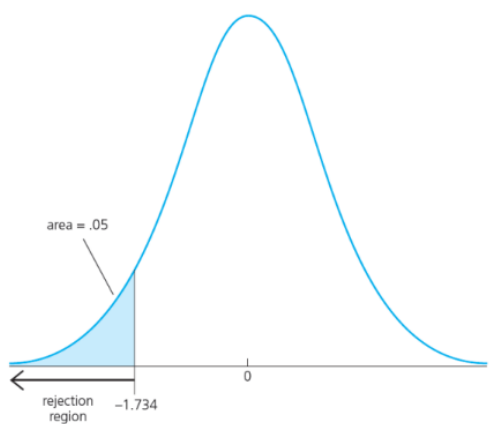


Figure 3.3: For example, if the significance level is 5% and the degrees of freedom is 18, then c=1.734, and so is rejected in favor of at the 5% level if $t_{\hat{B}}j$ < -1.734

3.1.2 p-Value Approach

The **p-value** is the smallest significance level α at which H_0 would be rejected given the data.

Decision rule:

$$\text{Reject } H_0 \text{ if } p\text{-value} < \alpha$$

Advantages:

- Avoids reliance on arbitrary cutoff values
- Provides more information than a binary accept/reject rule

3.1.3 Confidence Intervals for Regression Coefficients

A $(1-\alpha) \times 100\%$ confidence interval for β_j is given by:

$$\hat{\beta}_j \pm t_{n-k-1,1-\alpha/2} \cdot \mathrm{SE}(\hat{\beta}_j)$$

Interpretation: With repeated random sampling, this interval will contain the true parameter β_j in approximately $(1-\alpha) \times 100\%$ of samples.

Example: For $\alpha = 0.05$, a 95% confidence interval is:

$$\hat{\beta}_j \pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_j)$$

if $n-k-1$ is large (normal approximation).

Note: The interval becomes narrower when:

- n increases (more information)
- $\mathrm{SE}(\hat{\beta}_j)$ decreases (more precision)

3.1.4 Testing Linear Combinations of Coefficients

Often we are interested in testing hypotheses of the form:

$$H_0 : \beta_j - \beta_r = 0 \quad \text{vs.} \quad H_1 : \beta_j - \beta_r \neq 0$$

This is common when comparing the effect of two variables.

We define the linear combination:

$$\lambda = \beta_j - \beta_r \quad \text{and estimate it by} \quad \hat{\lambda} = \hat{\beta}_j - \hat{\beta}_r$$

Then we compute:

$$t = \frac{\hat{\lambda} - \lambda_0}{\mathrm{SE}(\hat{\lambda})} \quad \text{where } \lambda_0 = 0 \text{ under } H_0$$

To get $\mathrm{SE}(\hat{\lambda})$, we use:

$$\mathrm{Var}(\hat{\lambda}) = \mathrm{Var}(\hat{\beta}_j) + \mathrm{Var}(\hat{\beta}_r) - 2 \cdot \mathrm{Cov}(\hat{\beta}_j, \hat{\beta}_r)$$

Note: This requires the full variance–covariance matrix of the OLS estimator.

3.2 Joint Hypothesis Testing: The F-Test

The t-test handles single restrictions. To test **multiple restrictions jointly**, we use the F-test.

Null Hypothesis with q Restrictions

$$H_0 : \beta_{j_1} = 0,\ \beta_{j_2} = 0, \ldots, \beta_{j_q} = 0 \quad \text{(q restrictions)}$$

Procedure:

1. Estimate the unrestricted model $\rightarrow \text{SSR}_{\text{ur}}$
2. Estimate the restricted model (imposing H_0) $\rightarrow \text{SSR}_{\text{r}}$
3. Compute the F-statistic:

$$F = \frac{(\text{SSR}_{\text{r}} - \text{SSR}_{\text{ur}})/q}{\text{SSR}_{\text{ur}}/(n-k-1)}$$

Decision Rule:

$$\text{Reject } H_0 \text{ if } F > F_{q,n-k-1,\alpha}$$

The F-distribution has q and $n-k-1$ degrees of freedom.

Interpretation

- If the restrictions hold (i.e., H_0 is true), the restricted model should not fit much worse than the unrestricted model
- A large F-statistic indicates that the restrictions are invalid $\rightarrow$ reject H_0

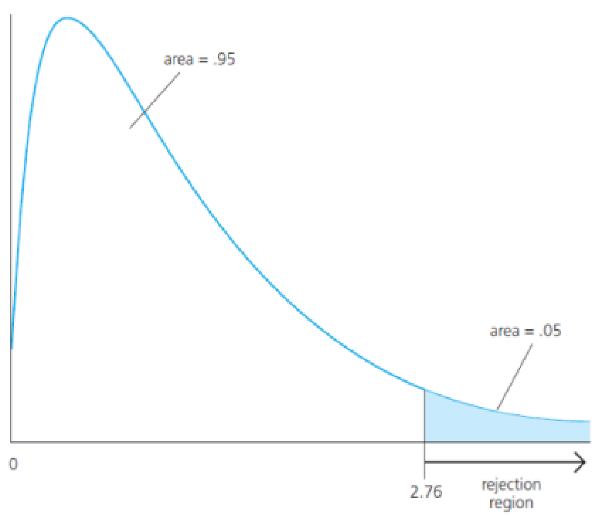


Figure 3.4: We will reject H0 in favor of H1 when F is sufficiently "large." How large depends on our chosen significance level. Suppose that we have decided on a 5% level test. Let c be the 95th percentile in the $F_q, n-k-1$ distribution. Once c has been obtained, we reject H0 in favor of H1 at the chosen significance level if F¿c

Alternative Formula for the F Statistic (Using R^2)

If the dependent variable is the same across the restricted and unrestricted models, the F statistic can also be computed using the R^2 values:

$$F = \frac{(R^2_{ur} - R^2_r)/q}{(1 - R^2_{ur})/(n - k - 1)} = \frac{(R^2_{ur} - R^2_r)/q}{(1 - R^2_{ur})/df_{ur}}$$

where:

- R^2_{ur}: R^2 from the unrestricted model
- R^2_r: R^2 from the restricted model
- q: number of restrictions being tested
- $df_{ur} = n - k - 1$: degrees of freedom in the unrestricted model

Interpretation: This version expresses the F statistic as a scaled difference in goodness-of-fit. A large drop in R^2 (relative to degrees of freedom lost) implies the restrictions significantly reduce the model's explanatory power.

Important: This formula is valid only when:

- The restricted and unrestricted models have the same dependent variable
- The models are nested

It should **not** be used to compare models with different dependent variables.

3.2.1 Lagrange Multiplier (LM) Test

The **Lagrange Multiplier (LM) test**, also called the **score test**, provides an alternative method for testing multiple linear restrictions.

Idea Behind the LM Test

Instead of estimating the unrestricted model (as in the F-test), the LM test only requires estimation of the restricted model.

Motivation: When the unrestricted model is hard to estimate or computationally intensive, the LM test is useful because it avoids full re-estimation.

Typical Application

Suppose we want to test:

$$H_0 : \beta_{k-q+1} = \beta_{k-q+2} = \cdots = \beta_k = 0$$

Instead of computing the unrestricted regression, we:

1. Estimate the **restricted model**
2. Use the residuals to evaluate whether the omitted variables (the ones restricted to zero) significantly explain variation in the dependent variable

Test Statistic and Distribution

The LM statistic asymptotically follows:

$$LM \sim \chi^2_q$$

where q is the number of restrictions being tested.

Decision Rule

Reject H_0 at significance level α if:

$$LM > \chi^2_{q,1-\alpha}$$

Note: Under homoskedasticity and normality, the LM test and F-test are asymptotically equivalent.

When to Use LM

- Useful when the restricted model is much simpler than the unrestricted one
- Common in large-scale models or systems of equations
- More efficient in large samples when full re-estimation is costly

Chapter 4

Lecture 5

4.1 Heteroskedasticity in the Multiple Regression Model

What Is Heteroskedasticity?

We consider the usual multiple linear regression model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + u_i$$

Under assumptions MLR.1–MLR.4, the OLS estimators remain **unbiased**, even if the error variance changes across observations.

However, when the assumption MLR.5 (homoskedasticity) is violated, we say the model suffers from **heteroskedasticity**:

$$\text{Var}(u_i \mid x_i) = \sigma_i^2 \quad \text{(not constant)}$$

Key Consequences of Heteroskedasticity

- **Unbiasedness:** OLS estimators are still unbiased under MLR.1–MLR.4
- **Efficiency:** OLS is no longer BLUE — it is not the most efficient linear estimator
- **Inference:** Standard errors, t statistics, and confidence intervals based on homoskedasticity are **invalid**

Goodness-of-fit (e.g., R^2) is unaffected by heteroskedasticity, since it depends only on unconditional variances.

4.2 OLS Variance and Robust Estimation

Variance of OLS with Heteroskedastic Errors

Even though the OLS estimator remains:

$$\hat{\beta}_1 = \beta_1 + \frac{\sum(x_i - \bar{x})y_i}{\sum(x_i - \bar{x})^2}$$

its sampling variance is no longer:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum(x_i - \bar{x})^2}$$

Instead:

$$\text{Var}(\hat{\beta}_1) = \frac{\sum(x_i - \bar{x})^2 \cdot \sigma_i^2}{\left(\sum(x_i - \bar{x})^2\right)^2}$$

where $\left(\sum(x_i - \bar{x})^2\right)^2 = SST^2$

Conclusion: The homoskedastic formula for standard errors is no longer valid.

4.2.1 White's Heteroskedasticity-Robust Standard Errors (1980)

White (1980) proposed a consistent estimator of the variance of $\hat{\beta}_j$, robust to arbitrary forms of heteroskedasticity.

In the simple case:

$$\widehat{\text{Var}}(\hat{\beta}_1) = \frac{\sum(x_i - \bar{x})^2 \cdot \hat{u}_i^2}{SST^2}$$

In the general MLR case, robust variance is:

$$\widehat{\text{Var}}(\hat{\beta}_j) = \frac{\sum \hat{u}_i^2 r_{ji}^2}{SSR_j^2}$$

where r_{ji} are the residuals from regressing x_j on all other x's.

Asymptotic Justification: When this expression is multiplied by the sample size n, it converges in probability to:

$$\mathbb{E}\left[(x_i - \mu_x)^2 u_i^2\right] / (\sigma_x^2)^2$$

which is the probability limit of $n \cdot \text{Var}(\hat{\beta}_1)$, as shown in the population formula for heteroskedastic variance. The **heteroskedasticity-robust standard error** (also called White-Huber-Eicker SE) is:

$$\text{SE}_{\text{robust}}(\hat{\beta}_j) = \sqrt{\widehat{\text{Var}}(\hat{\beta}_j)}$$

Note: Sometimes a degrees-of-freedom adjustment $\frac{n}{n-k-1}$ is applied.

Why? If all $\hat{u}_i^2$ were equal (strongest possible form of homoskedasticity), then the robust formula would reduce exactly to the usual homoskedastic standard error. This correction ensures consistency with the homoskedastic case.

When Should We Use Robust Standard Errors?

General rule: Always use robust standard errors unless you have strong evidence that the error variance is homoskedastic.

- They are consistent whether or not heteroskedasticity is present
- Under homoskedasticity, robust SEs will be *asymptotically equivalent* to regular SEs
- Under heteroskedasticity, regular SEs are **biased and inconsistent**

Therefore: Robust SEs are a safe and general default for valid inference.

4.3 Testing for Heteroskedasticity

Setup and Null Hypothesis

We consider the multiple linear regression model under assumptions MLR.1 through MLR.4. In particular:

$$\mathbb{E}[u_i \mid x_i] = 0 \quad \text{and} \quad \mathrm{Var}(u_i \mid x_i) = \sigma_i^2$$

The hypothesis of homoskedasticity (Assumption MLR.5) is:

$$H_0 : \mathrm{Var}(u_i \mid x_i) = \sigma^2 \quad \text{(constant)}$$

We want to test whether this assumption is violated — that is, whether the variance of the errors depends on the values of the regressors.

Intuition

Under the alternative, the variance of u_i is systematically related to one or more regressors.

We can model this by assuming that:

$$\mathbb{E}[u_i^2 \mid x_i] = h(x_i)$$

A simple functional form is to assume that $\mathbb{E}[u_i^2 \mid x_i]$ is linearly related to the regressors:

$$\mathbb{E}[u_i^2 \mid x_i] = \gamma_0 + \gamma_1 x_{1i} + \cdots + \gamma_k x_{ki}$$

Then, the null hypothesis becomes:

$$H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_k = 0$$

This setup allows us to use an F test or LM (Lagrange Multiplier) test on this auxiliary regression.

Auxiliary Regression

We estimate the equation:

$$\hat{u}_i^2 = \gamma_0 + \gamma_1 x_{1i} + \cdots + \gamma_k x_{ki} + v_i$$

and use the R^2 from this regression to construct an F-statistic for testing the exclusion restrictions:

$$F = \frac{R^2/k}{(1-R^2)/(n-k-1)} \quad \text{(approximately follows an } F_{k,n-k-1} \text{ under } H_0\text{)}$$

LM Statistic

An alternative version is the Breusch–Pagan (LM) test:

$$LM = n \cdot R^2$$

Where R^2 is from the auxiliary regression of $\hat{u}_i^2$ on the regressors.
Under H_0, the LM statistic follows:

$$LM \sim \chi_k^2$$

Summary: Both the F-statistic and the LM-statistic test whether the error variance depends on the regressors. The LM test is more commonly reported and is asymptotically equivalent to the F-test in large samples.

4.4 White Test for Heteroskedasticity

The **White test** is a more general test for heteroskedasticity than the Breusch–Pagan test, because it allows the error variance to depend on:

- the regressors, their squares, and their cross-products.

Auxiliary Regression Equation

After estimating the OLS model and obtaining residuals $\hat{u}_i$, we run the following auxiliary regression:

$$\hat{u}_i^2 = \gamma_0 + \gamma_1 x_{1i} + \gamma_2 x_{2i} + \gamma_3 x_{1i}^2 + \gamma_4 x_{2i}^2 + \gamma_5 x_{1i} x_{2i} + \cdots + v_i$$

This includes:

- All the original regressors, The squares of regressors and All unique pairwise products (interaction terms)

Let R^2_{aux} be the R^2 from the auxiliary regression above. The **White test statistic** is:

$$LM = n \cdot R^2_{\text{aux}}$$

Under the null hypothesis of homoskedasticity, the **Decision rule** is:

$$LM \sim \chi_q^2 \quad \text{where } q = \text{number of terms in the auxiliary regression excluding the constant}$$

Reject H_0 if:

$$LM > \chi^2_{q,1-\alpha}$$

Interpretation

The White test allows a flexible form for heteroskedasticity and is consistent against many alternatives. However, it has the drawback of involving an abundance of regressors, thus using many degrees of freedom for models with just a moderate number of independent variables. This means it also lose power where the sample size is small

4.5 Correcting for Heteroskedasticity

4.5.1 Weighted Least Squares (WLS)

Why OLS Fails Under Heteroskedasticity

Recall the usual OLS objective:

$$\min_{\beta_0,\ldots,\beta_k} \sum_{i=1}^{n} \hat{u}_i^2$$

Problem: If the error variance is not constant — i.e., $\text{Var}(u_i \mid x_i) = \sigma_i^2$ varies across i — then this minimization gives **equal weight** to all residuals, even if some observations are much noisier than others.

This means:

- High-variance observations distort the OLS estimation
- OLS loses efficiency: it is no longer the Best Linear Unbiased Estimator (BLUE)

WLS: Correcting for Unequal Variances

Suppose the form of heteroskedasticity is known:

$$\text{Var}(u_i \mid x_i) = \sigma_i^2 \quad \text{with known } \sigma_i^2 > 0$$

To restore homoskedasticity, we transform the model by dividing all terms by σ_i:

$$\frac{y_i}{\sigma_i} = \frac{\beta_0}{\sigma_i} + \frac{\beta_1 x_{1i}}{\sigma_i} + \cdots + \frac{\beta_k x_{ki}}{\sigma_i} + \frac{u_i}{\sigma_i}$$

In the transformed model:

$$\text{Var}\left(\frac{u_i}{\sigma_i}\right) = 1 \quad \Rightarrow \quad \text{homoskedasticity is restored}$$

WLS Estimation Procedure

WLS minimizes the weighted sum of squared residuals:

$$\min_{\beta_0,\ldots,\beta_k} \sum_{i=1}^{n} w_i \cdot \hat{u}_i^2 \quad \text{where } w_i = \frac{1}{\sigma_i^2}$$

Intuition:

- Observations with higher variance get lower weight
- Observations with more reliable information (lower variance) get more influence

Result: WLS produces estimators that are:

- **Unbiased**, provided that the model is correctly specified
- **Efficient** — they are BLUE under heteroskedasticity with known variance

4.5.2 Feasible Generalized Least Squares (FGLS)

WLS requires knowledge of the σ_i^2, which is rarely available in practice — motivating the use of *Feasible GLS (FGLS)*, where we estimate those variances from the data.

- Model the error variance $h(x_i)$
- Estimate it from the data
- Use those estimates to transform the regression and apply Weighted Least Squares

Modeling the Error Variance Function

We assume a flexible parametric form for the conditional variance:

$$\text{Var}(u_i \mid x_i) = \sigma^2 \exp(\delta_0 + \delta_1 x_{1i} + \delta_2 x_{2i} + \cdots + \delta_k x_{ki})$$

The exponential form ensures the predicted variances are always positive (unlike linear forms such as in the Breusch–Pagan setup).

Deriving a Linear Estimable Equation

Start from:

$$u_i^2 = \sigma^2 \exp(\delta_0 + \delta_1 x_{1i} + \cdots + \delta_k x_{ki}) \cdot v_i$$

where v_i is a random variable with:

$$\mathbb{E}[v_i \mid x_i] = 1$$

Taking logs:

$$\log(u_i^2) = a_0 + \delta_1 x_{1i} + \delta_2 x_{2i} + \cdots + \delta_k x_{ki} + e_i$$

where e_i is a new error term with zero mean, independent of x_i.

Step-by-Step Estimation Procedure

1. Run an OLS regression, obtain residuals $\hat{u}_i$, and compute $\log(\hat{u}_i^2)$
2. Estimate the log-variance model:
$$\log(\hat{u}_i^2) = \hat{a}_0 + \hat{\delta}_1 x_{1i} + \cdots + \hat{\delta}_k x_{ki}$$
3. Compute the predicted values:
$$\hat{h}_i = \exp(\hat{a}_0 + \hat{\delta}_1 x_{1i} + \cdots + \hat{\delta}_k x_{ki})$$
4. Run WLS on the original regression model using weights:
$$w_i = \frac{1}{\hat{h}_i}$$

Chapter 5

Lecture 6

5.1 Regression Models for Qualitative Information

In many empirical applications, we want to model a **qualitative** outcome — especially binary events (e.g., employment status, loan default, purchase decision).

Let:

$$y_i \in \{0, 1\} \quad \text{(binary dependent variable)}$$

Examples:

- $y_i = 1$: the person is employed; $y_i = 0$: unemployed
- $y_i = 1$: a customer buys; $y_i = 0$: does not buy

5.1.1 The Linear Probability Model (LPM)

We can write a standard multiple linear regression:

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + u_i$$

But when $y \in \{0, 1\}$, interpretation changes:

- y no longer represents a continuous variable
- Instead, $\mathbb{E}[y \mid x] = P(y = 1 \mid x)$

This leads to the so-called **Linear Probability Model (LPM)**, where:

$$P(y = 1 \mid x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$$

Each β_j now measures the *change in the probability* that $y = 1$, associated with a one-unit increase in x_j, holding all else constant.

Strengths and Limitations of the Linear Probability Model

Advantages:

- Easy to estimate using OLS
- Coefficients are straightforward to interpret as marginal effects on probabilities
- Provides a reasonable approximation when most predicted probabilities stay near the center of the unit interval

Main Limitations:

- The model can predict probabilities outside the valid range $[0, 1]$
- The probability of success cannot be linear in x over the full domain — this leads to functional form misspecification

Even with these issues, the LPM is often useful in applied settings and works well when the values of the regressors are near their means in the sample.

Predicted Classification and Fit

Although predicted values from the LPM can lie outside $[0, 1]$, we can still define binary predictions by thresholding:

$$\hat{y}_i = \begin{cases} 1 & \text{if } \hat{p}_i \geq 0.5 \\ \\ 0 & \text{if } \hat{p}_i < 0.5 \end{cases}$$

We can then compare predicted vs. actual values to compute:

- Percentage of 1's correctly predicted
- Percentage of 0's correctly predicted
- Overall percent correctly predicted

This percentage is commonly used as a goodness-of-fit measure for binary dependent variable models.

Heteroskedasticity in the LPM

In an LPM, the error term is:

$$u_i = y_i - P(y_i = 1 \mid x_i)$$

Since $y_i \in \{0, 1\}$, we have:

$$\text{Var}(u_i \mid x_i) = P(y_i = 1 \mid x_i) \cdot [1 - P(y_i = 1 \mid x_i)] = p(x_i)(1 - p(x_i))$$

Conclusion: Unless $p(x_i)$ is constant across observations, the error variance depends on x_i, implying **heteroskedasticity**.

Implications:

- OLS estimates remain unbiased under MLR.1–MLR.4
- But the usual standard errors, t-statistics, and F-tests are invalid
- In practice, heteroskedasticity-robust standard errors should be used

5.2 Logit and Probit Models

The Linear Probability Model has several drawbacks, especially the possibility of predicting probabilities outside the $[0, 1]$ interval. To solve this, we consider models where the response probability is a nonlinear function of the regressors.

Binary Response Model: General Form

We assume:

$$P(y = 1 \mid x) = G(x\beta)$$

where:

- $x\beta = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$
- $G(z)$ is a strictly increasing function such that $0 < G(z) < 1$ for all real z

Two common choices for G:

- **Logit model:** $G(z) = \Lambda(z) = \frac{1}{1+\exp(-z)}$ This is the cumulative distribution function for a standard logistic random variable.
- **Probit model:** $G(z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$ This is the standard normal cumulative distribution function(cdf).

Both functions are:

- Sigmoid (S-shaped)
- Strictly increasing
- Ensure predicted probabilities lie in $(0, 1)$

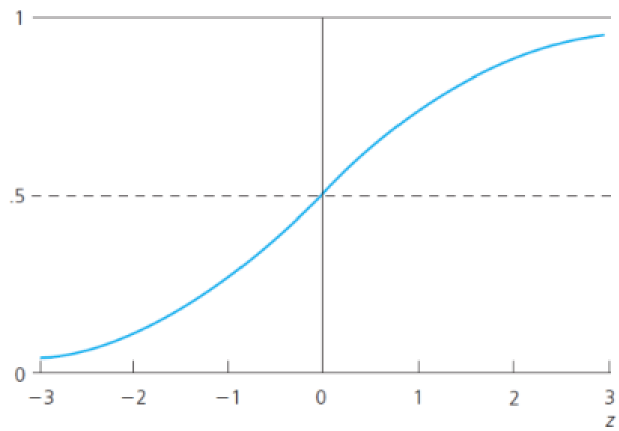


Figure 5.1: The standard normal cdf has a shape very similar to that of the logistic cdf.

5.2.1 Partial Effects of Continuous Variables

In both logit and probit models, the probability of success is:

$$p(\mathbf{x}) = G(\beta_0 + \mathbf{x}\beta)$$

The partial effect of a continuous variable x_j on $p(\mathbf{x})$ is:

$$\frac{\partial p(\mathbf{x})}{\partial x_j} = g(\beta_0 + \mathbf{x}\beta) \cdot \beta_j$$

where $g(z) = G'(z)$ is the density function associated with the CDF $G(z)$.

For each model:

- Probit: $g(z) = \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$
- Logit: $g(z) = \Lambda(z)(1 - \Lambda(z))$

Key Properties of the Partial Effects

- The partial effect of x_j always has the **same sign as** β_j
- The **magnitude** depends on $\mathbf{x}$ through the term $g(\beta_0 + \mathbf{x}\beta)$
- The **ratio** of partial effects for two regressors x_j and x_h is:

$$\frac{\partial p/\partial x_j}{\partial p/\partial x_h} = \frac{\beta_j}{\beta_h} \quad \text{(independent of } \mathbf{x}\text{)}$$

- The **maximum marginal effect** occurs when $\beta_0 + \mathbf{x}\beta = 0$, since $g(z)$ reaches its peak at $z = 0$

Numerical Benchmarks:

- Probit: $g(0) = \phi(0) = \frac{1}{\sqrt{2\pi}} \approx 0.40$
- Logit: $g(0) = \Lambda(0)[1 - \Lambda(0)] = \frac{1}{4} = 0.25$

5.3 Maximum Likelihood Estimation

Why OLS Is Inappropriate

In binary response models, the dependent variable is $y_i \in \{0,1\}$, and the model specifies:

$$P(y_i = 1 \mid \mathbf{x}_i) = G(\beta_0 + \mathbf{x}_i\beta)$$

OLS is no longer valid because:

- The conditional expectation $\mathbb{E}[y_i \mid \mathbf{x}_i]$ is nonlinear in the parameters
- The error term $u_i = y_i - G(\beta_0 + \mathbf{x}_i\beta)$ is heteroskedastic and non-normal

Instead, we use **Maximum Likelihood Estimation (MLE)** to estimate the parameters.

5.3.1 Log-Likelihood Function

Given a binary model with $y_i \in \{0,1\}$, and success probability:

$$P(y_i = 1 \mid \mathbf{x}_i) = G(\beta_0 + \mathbf{x}_i\beta)$$

The contribution of observation i to the likelihood is:

$$\mathcal{L}_i(\theta) = [G(\beta_0 + \mathbf{x}_i\beta)]^{y_i} \cdot [1 - G(\beta_0 + \mathbf{x}_i\beta)]^{1-y_i}$$

the log-likelihood function for the full sample of size n is:

$$\mathcal{L}(\beta) = \sum_{i=1}^{n} \ell_i(\beta) \quad \text{where } \ell_i(\beta) = y_i \log G(\beta_0 + \mathbf{x}_i\beta) + (1 - y_i)\log(1 - G(\beta_0 + \mathbf{x}_i\beta))$$

Maximum Likelihood Estimator: The estimator $\hat{\beta}$ is the value that maximizes $\mathcal{L}(\beta)$. It satisfies:

$$\hat{\beta} = \arg\max_{\beta} \mathcal{L}(\beta)$$

Model-specific naming:

- If $G(\cdot) = \Lambda(\cdot) \to \hat{\beta}$ is the *logit estimator*
- If $G(\cdot) = \Phi(\cdot) \to \hat{\beta}$ is the *probit estimator*

Remark: Due to the nonlinearity of $G(\cdot)$, there is no closed-form solution. Computation requires numerical optimization (e.g., Newton-Raphson, BFGS).

Nevertheless, MLE theory ensures:

- $\hat{\beta}$ is **consistent**, **asymptotically normal**, **asymptotically efficient**

5.3.2 Inference and Hypothesis Testing

Each estimated coefficient $\hat{\beta}_j$ has an asymptotic standard error $\text{se}(\hat{\beta}_j)$, which is obtained from the estimated information matrix.

Once we have standard errors, we can:

- Construct confidence intervals:

$$\hat{\beta}_j \pm z_{1-\alpha/2} \cdot \text{se}(\hat{\beta}_j)$$

- Conduct hypothesis testing. To test:

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0$$

  use the asymptotic t-statistic:

$$z_j = \frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)} \quad \text{and compare to standard normal quantiles}$$

Testing multiple restrictions:

To jointly test q restrictions on the model (e.g., $H_0 : \beta_1 = \beta_2 = 0$), we have two classical options:

- **Lagrange Multiplier (LM) Test:** Based on score function evaluated at restricted estimates.

- **Likelihood Ratio (LR) Test:** Based on comparison of log-likelihoods:

$$LR = 2\left[\mathcal{L}(\hat{\beta}_{ur}) - \mathcal{L}(\hat{\beta}_r)\right] \quad \sim \chi^2_q \text{ under } H_0$$

  where $\hat{\beta}_{ur}$ is the unrestricted MLE, and $\hat{\beta}_r$ is the MLE under the null.

Asymptotic properties guarantee: These tests are valid in large samples.

Chapter 6

Lecture 7

6.1 Instrumental Variables Estimation

Endogeneity and Omitted Variable Bias

Endogeneity often arises due to **omitted variables**. In such cases, we cannot assume that the explanatory variable is exogenous — that is, uncorrelated with the error term. This leads us to two alternatives:

- **Ignore** the omitted variable and accept the bias in OLS estimates
- Attempt to find a **proxy variable** for the omitted factor

When no valid proxy is available, we must resort to a method that explicitly addresses endogeneity: **Instrumental Variables (IV) estimation**.

Example: In a wage equation, ability is unobserved but likely correlated with education. If we omit ability, we estimate:

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \cdot \text{educ}_i + u_i, \quad \text{where } u_i = \text{abil}_i + \text{other unobservables}$$

If educ_i and abil_i are correlated, then educ_i is endogenous and the OLS estimator of β_1 is biased.

6.1.1 Instrumental Variables and the Moment Condition

Consider the model:

$$y_i = \beta_0 + \beta_1 x_i + u_i \quad \text{with } \text{Cov}(x_i, u_i) \neq 0$$

Suppose we observe an instrument z_i such that:

- **Relevance:** $\text{Cov}(z_i, x_i) \neq 0$
- **Exogeneity:** $\text{Cov}(z_i, u_i) = 0$

Then, we can identify β_1 through the following moment condition:

$$\mathbb{E}[z_i y_i] = \beta_1 \mathbb{E}[z_i x_i] \quad \Rightarrow \quad \beta_1 = \frac{\text{Cov}(z_i, y_i)}{\text{Cov}(z_i, x_i)}$$

6.1.2 Sample IV Estimator

Given a sample $\{(y_i, x_i, z_i)\}_{i=1}^n$, we estimate β_1 as:

$$\hat{\beta}_1^{IV} = \frac{\sum_{i=1}^n (z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^n (z_i - \bar{z})(x_i - \bar{x})}$$

Properties:

- If $z_i = x_i$, then $\hat{\beta}_1^{IV} = \hat{\beta}_1^{OLS}$
- If the IV assumptions hold, $\hat{\beta}_1^{IV}$ is consistent
- In small samples, $\hat{\beta}_1^{IV}$ is generally not unbiased

The Problem of Weak Instruments

The asymptotic variance of the IV estimator under homoskedasticity is:

$$\text{Var}(\hat{\beta}_1^{IV}) \approx \frac{\sigma_u^2}{\sigma_x^2 \cdot \rho_{xz}^2}$$

Where:

- σ_u^2: variance of the error term
- σ_x^2: variance of the endogenous regressor x_i
- ρ_{xz}: correlation between x_i and z_i

If ρ_{xz} is small (i.e., the instrument is weak), the variance of $\hat{\beta}_1^{IV}$ becomes large, and inference becomes unreliable. In such cases:

- The sampling distribution of the estimator becomes heavily skewed
- The asymptotic normal approximation fails
- Standard inference tools like t-tests and confidence intervals are invalid

Instrumental Variables with Additional Covariates

Now consider a model with multiple regressors:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + u_i$$

Suppose:

- x_{1i} is endogenous
- $x_{2i}, \ldots, x_{ki}$ are exogenous

- z_i is a valid instrument for x_{1i}

The key IV moment condition becomes:

$$\mathbb{E}[z_i \cdot (y_i - \beta_0 - \beta_1 x_{1i} - \beta_2 x_{2i} - \cdots - \beta_k x_{ki})] = 0$$

Thus, we can consistently estimate β_1 by:

- Controlling for the exogenous regressors in both stages
- Instrumenting only the endogenous regressor with z_i

Causal Interpretation of the IV Estimator

The motivation for IV estimation is to recover the causal effect of x on y even in the presence of omitted variable bias or simultaneity.

By isolating variation in x that is induced by the exogenous instrument z, the IV estimator effectively removes the endogenous part of x. As long as:

- z is strongly correlated with x, and
- z is uncorrelated with the structural error u

then:

$$\hat{\beta}_1^{IV} \longrightarrow \beta_1 \quad \text{(consistently)}$$

This justifies interpreting $\hat{\beta}_1^{IV}$ as a causal parameter.

6.1.3 From Endogeneity to Instrumental Variables

Suppose we are interested in estimating the structural equation:

$$y_{1i} = \beta_0 + \beta_1 y_{2i} + \beta_2 z_{1i} + u_{1i}$$

where y_2 is potentially endogenous — that is, correlated with the error term u_1. In this case, the OLS estimator of β_1 will be biased and inconsistent. The core problem is that:

$$\mathbb{E}(y_2 u_1) \neq 0$$

To address this, we introduce one or more **instrumental variables** (IVs): variables that are correlated with the endogenous regressor but uncorrelated with the structural error. These must satisfy two key conditions:

- **Relevance:** $\text{Cov}(z_j, y_2) \neq 0$
- **Exogeneity:** $\text{Cov}(z_j, u_1) = 0$

However, if an instrument like z_1 is already included as a regressor, it cannot serve as an IV. We need **excluded exogenous variables** — e.g., z_2 and z_3 — that do not appear in the structural equation and satisfy the IV conditions.

Moment Conditions and Identification

The IV estimator can be derived using the method of moments. If we assume that z_1, z_2, z_3 are uncorrelated with the error, then valid moment conditions are:

$$\sum_{i=1}^{n} z_{j,i}(y_{1i} - \beta_0 - \beta_1 y_{2i} - \beta_2 z_{1i}) = 0 \quad \text{for } j = 1, 2, 3$$

These equations form the basis for estimating $\beta_0, \beta_1, \beta_2$. However, using each instrument individually leads to different estimators — none of which is generally efficient. We need a systematic method to combine all instruments.

6.2 Two-Stage Least Squares (2SLS)

6.2.1 The Reduced Form: First-Stage Regression

To construct a single best IV for y_2, we consider its reduced form:

$$y_{2i} = \pi_0 + \pi_1 z_{1i} + \pi_2 z_{2i} + \pi_3 z_{3i} + v_{2i}$$

This regression decomposes y_2 into:

- $y_2^* = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + \pi_3 z_3$: the part of y_2 explained by exogenous instruments (and thus uncorrelated with u_1)
- v_2: the residual, potentially correlated with u_1

The key identification condition is:

$$\pi_2 \neq 0 \quad \text{or} \quad \pi_3 \neq 0$$

Without this, y_2 is not sufficiently explained by the excluded instruments, and the model is underidentified.

6.2.2 Two-Stage Least Squares Procedure

Stage 1: Run the reduced form regression:

$$y_{2i} = \pi_0 + \pi_1 z_{1i} + \pi_2 z_{2i} + \pi_3 z_{3i} + v_{2i}$$

Obtain fitted values:

$$\hat{y}_{2i} = \hat{\pi}_0 + \hat{\pi}_1 z_{1i} + \hat{\pi}_2 z_{2i} + \hat{\pi}_3 z_{3i}$$

Stage 2: Use $\hat{y}_{2i}$ in place of y_{2i} in the structural equation and estimate via OLS:

$$y_{1i} = \beta_0 + \beta_1 \hat{y}_{2i} + \beta_2 z_{1i} + \varepsilon_i$$

This process gives the 2SLS estimator $\hat{\beta}_1^{2SLS}$, which is consistent and asymptotically normal under standard assumptions.

Why 2SLS Works

We can rewrite the model as:

$$y_2 = y_2^* + v_2 \quad \Rightarrow \quad y_1 = \beta_0 + \beta_1 y_2^* + \beta_1 v_2 + \beta_2 z_1 + u_1$$

Then:

$$y_1 = \beta_0 + \beta_1 y_2^* + \beta_2 z_1 + \underbrace{(u_1 + \beta_1 v_2)}_{\text{composite error}}$$

Since y_2^* and z_1 are uncorrelated with the composite error, the OLS assumptions are satisfied.

Testing Instrument Strength

A necessary step is to verify that z_2 and z_3 are jointly significant predictors of y_2 in the first-stage regression. Weak instruments lead to:

- High variance of 2SLS estimators
- Invalid inference due to weak identification

Variance and Multicollinearity Issues

2SLS estimators tend to be less efficient than OLS for two main reasons:

1. $\hat{y}_2$ has less variance than y_2, being a fitted value
2. $\hat{y}_2$ may be highly collinear with other regressors like z_1

Identification Conditions

- **Order condition (necessary):** The number of excluded instruments must be at least as large as the number of endogenous regressors.
- **Rank condition (sufficient):** The excluded instruments must have sufficient partial correlation with the endogenous variables, conditional on the included exogenous variables.

If these conditions are not satisfied — for example, if one instrument appears in both reduced forms but the other appears in neither — 2SLS will fail to produce consistent estimators.

Multiple Endogenous Regressors

Now consider a model with two endogenous regressors:

$$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 y_3 + \beta_3 z_1 + u_1$$

To identify the model via 2SLS, we need at least two valid instruments excluded from the equation, say z_4 and z_5. These instruments must appear with nonzero coefficients in the reduced forms:

$$y_2 = \pi_0 + \pi_1 z_1 + \pi_2 z_4 + \pi_3 z_5 + v_2$$

$$y_3 = \gamma_0 + \gamma_1 z_1 + \gamma_2 z_4 + \gamma_3 z_5 + v_3$$

6.3 Testing for Endogeneity

IV/2SLS should only be used when endogeneity is a concern. But how do we test for it?

Hausman (1978) proposed a test that compares the OLS and 2SLS estimators. Under the null hypothesis that y_2 is exogenous:

$$\hat{\beta}_{OLS} \text{ and } \hat{\beta}_{2SLS} \text{ are both consistent, but } \hat{\beta}_{OLS} \text{ is efficient}$$

$$H_0 : \text{OLS is consistent and efficient} \quad \text{vs.} \quad H_1 : \text{OLS is inconsistent}$$

Thus, large differences between the two estimators signal endogeneity.

The Hausman Test

To illustrate the idea, suppose we estimate the following structural equation:

$$y_{1i} = \beta_0 + \beta_1 y_{2i} + \beta_2 z_{1i} + \beta_3 z_{2i} + u_{1i}$$

We suspect that y_2 might be endogenous. We have two instruments, z_3 and z_4, which are excluded exogenous variables. If y_2 is in fact uncorrelated with u_1, then we should use OLS. How can we test this?

We begin by estimating the reduced-form equation for y_2:

$$y_{2i} = \pi_0 + \pi_1 z_{1i} + \pi_2 z_{2i} + \pi_3 z_{3i} + \pi_4 z_{4i} + v_{2i}$$

Since the instruments are uncorrelated with u_1, the correlation between y_2 and u_1 depends entirely on whether v_2 is correlated with u_1. This motivates the regression-based test.

Assume the structural error can be written as:

$$u_{1i} = \delta_1 v_{2i} + e_i, \quad \text{where } \mathbb{E}(e_i \mid v_{2i}) = 0$$

Under the null hypothesis of exogeneity, we have $\delta_1 = 0$. We cannot observe v_2, but we can estimate it from the first-stage residuals:

$$\hat{v}_2 = y_2 - \hat{y}_2$$

We then estimate the augmented regression:

$$y_{1i} = \beta_0 + \beta_1 y_{2i} + \beta_2 z_{1i} + \beta_3 z_{2i} + \delta_1 \hat{v}_{2i} + \varepsilon_i$$

A simple t-test on $\hat{\delta}_1$ allows us to test for endogeneity. If $\hat{\delta}_1$ is significantly different from zero, we reject the null and conclude that y_2 is endogenous.

Test $H_0 : \delta = 0$. If rejected, conclude that y_2 is endogenous

6.4 Some Applications

Returns to Education

$$\text{Education} \rightarrow \text{Wage}$$

IV: Mother's education (exogenous, affects child's education but not directly the wage).

Venture Capital and Start-Up Growth

$$\text{VC financing} \rightarrow \text{Start-up growth}$$

IVs: Local industry characteristics and geographic location (impact VC funding but not directly growth).

Family Succession and Firm Performance

$$\text{Family succession} \rightarrow \text{Firm performance}$$

IV: Gender of departing CEO's firstborn (affects succession decisions, not performance directly).

Political Protests and Policy Change

$$\text{Protest attendance} \rightarrow \text{Political change}$$

IV: Rainfall during protest days (exogenous factor reducing attendance).

These cases illustrate the power of IV methods in uncovering causal relationships where OLS would fail due to endogeneity.

Chapter 7

Lecture 8

7.1 Data over Time

7.1.1 Independently Pooled Cross Sections

An independently pooled cross section consists of samples drawn randomly from a population at different points in time. Each observation is independent, and the sampled individuals may differ across time. This setup is useful for analyzing population-level changes without tracking the same units over time.

Key Features:

- No unit is observed more than once.
- Observations are independent across time and across units.
- Useful for policy evaluation or studying trends when panel data is unavailable.

Estimation:

When analyzing pooled cross sections, we typically allow the intercept to differ across periods by including year dummies:

$$y_{it} = \beta_0 + \delta_t + \beta_1 x_{it} + u_{it}$$

Here, δ_t captures changes in the average outcome between years.

Interactions: We can also explore how the effect of x_{it} varies over time by interacting it with the time dummy:

$$y_{it} = \beta_0 + \delta_t + \beta_1 x_{it} + \beta_2 (x_{it} \cdot d_t) + u_{it}$$

This allows us to test for temporal heterogeneity in the effect of the explanatory variable.

Special case – two years: If we only have two periods (e.g., 2000 and 2005), interacting each regressor with a year dummy is equivalent to estimating separate regressions for each year.

Limitations:

- No control for unobserved individual-specific characteristics.

- Cannot assess dynamics within units.
- Assumes random sampling within each year.

7.1.2 Two-Period Panel Data

Panel data (also called longitudinal data) consists of repeated observations on the same units across multiple time periods. With two-period panel data, each unit (e.g., individual, firm, city) is observed at exactly two points in time, $t = 1$ and $t = 2$.

This setting allows us to control for *unobserved, time-invariant* characteristics by exploiting within-unit variation.

Motivation

Suppose we estimate:

$$y_{it} = \beta_0 + \beta_1 x_{it} + v_{it} \quad \text{with } v_{it} = a_i + u_{it}$$

where:

- a_i: unobserved, time-invariant effect (e.g., innate ability, firm culture),
- u_{it}: idiosyncratic, time-varying error.

If a_i is correlated with x_{it}, OLS gives a biased estimate of β_1 — this is called **heterogeneity bias**.

First-Differencing to Eliminate Fixed Effects

Because a_i is constant over time, differencing across periods removes it:

$$\begin{aligned} y_{i2} &= \beta_0 + \delta_0 + \beta_1 x_{i2} + a_i + u_{i2} \\ y_{i1} &= \beta_0 + \beta_1 x_{i1} + a_i + u_{i1} \end{aligned}$$

Subtracting:

$$\Delta y_i = y_{i2} - y_{i1} = \delta_0 + \beta_1 \Delta x_i + \Delta u_i$$

This is called the **first-differenced (FD) equation**, and the OLS estimate of β_1 from this equation is known as the **FD estimator**.

Assumptions for Valid Inference

To obtain a consistent estimate of β_1, two key assumptions are needed:

1. **Strict exogeneity**: The change in explanatory variables Δx_i must be uncorrelated with the change in error Δu_i.
2. **Sufficient variation**: Δx_i must vary across i; otherwise, we cannot identify the effect.

Why Not Just Pool?

Pooling the data and running OLS assumes that a_i is uncorrelated with x_{it}, which is often unrealistic. Moreover, the composite error $v_{it} = a_i + u_{it}$ introduces serial correlation:

$$\text{Cov}(v_{i1}, v_{i2}) = \text{Var}(a_i)$$

This violates the classical OLS assumptions and leads to:

- Biased standard errors
- Underestimated standard errors (if $a_i \neq 0$)

Practical Limitations of FD Estimation

- **Data Collection**: Following the same unit across two years is challenging, especially for individuals (e.g., due to non-response or dropout).
- **Loss of Variation**: Differencing reduces variation in x, which may weaken statistical power and increase standard errors.

7.1.3 Differencing with More Than Two Time Periods

When we observe more than two time periods per unit, we can still eliminate the unobserved effect a_i by differencing adjacent time periods. Consider the general fixed effects model:

$$y_{it} = \delta_1 + \delta_2 d_{2t} + \delta_3 d_{3t} + \beta_1 x_{it1} + \cdots + \beta_k x_{itk} + a_i + u_{it} \quad \text{for } t = 1, 2, 3$$

where d_{2t} and d_{3t} are time dummies. The total number of observations is $3N$.

The key assumption here is **strict exogeneity** of the explanatory variables, which requires:

$$\text{Cov}(x_{ij}, u_{is}) = 0 \quad \text{for all } t, s, j$$

This means that the explanatory variables must be uncorrelated with the idiosyncratic error in every period. This assumption is violated if we use lagged dependent variables or omit relevant time-varying covariates.

7.1.4 First-Differencing Over Multiple Periods

To remove the fixed effect a_i, we subtract adjacent periods. With $T = 3$, this yields:

$$\Delta y_{it} = \delta_2 \Delta d_{2t} + \delta_3 \Delta d_{3t} + \beta_1 \Delta x_{it1} + \cdots + \beta_k \Delta x_{itk} + \Delta u_{it} \quad \text{for } t = 2, 3$$

Note: $\Delta d_{2t} = 1$ for $t = 2$, and $\Delta d_{3t} = 1$ for $t = 3$, while the other dummy differences equal zero or negative one. Thus, the model includes no intercept by construction, which complicates the computation of statistics such as R^2.

To simplify interpretation and preserve an intercept, it is common to estimate:

$$\Delta y_{it} = \alpha_0 + \alpha_3 d_{3t} + \beta_1 \Delta x_{it1} + \cdots + \Delta u_{it} \quad \text{for } t = 2, 3$$

This allows us to account for time variation using a single period dummy (e.g., for period 3).

Extending to General $T > 3$

When $T > 3$, we refer to the dataset as a **balanced panel** if every unit is observed in every time period. After differencing, we generally include time dummies to capture secular trends:

$$\Delta y_{it} = \alpha_0 + \alpha_3 d_{3t} + \alpha_4 d_{4t} + \cdots + \alpha_T d_{Tt} + \beta_1 \Delta x_{it1} + \cdots + \Delta u_{it} \quad \text{for } t = 2, \ldots, T$$

Here, the total number of usable observations becomes $N(T-1)$, one per differenced pair.

Assumptions on Error Terms

For standard inference (e.g., valid t-tests), we must assume that the differenced error Δu_{it} is **not serially correlated**. However, this assumption is violated if the original error term u_{it} is serially correlated, even with constant variance. In that case, we typically observe:

$$\text{Corr}(\Delta u_{it}, \Delta u_{it-1}) = -0.5$$

Only when u_{it} follows a random walk will the differenced errors be uncorrelated.

Testing for Serial Correlation

Let $r_{it} = \Delta u_{it}$ be the differenced residuals from the first-differenced model. We test whether r_{it} follows an AR(1) process:

$$r_{it} = \rho r_{it-1} + e_{it}$$

The null hypothesis is $H_0 : \rho = 0$. The test proceeds as follows:

1. Estimate the first-differenced model and compute residuals $\hat{r}_{it}$
2. Regress $\hat{r}_{it}$ on $\hat{r}_{it-1}$ to estimate $\hat{\rho}$
3. Perform a t-test on $\hat{\rho}$ to test for serial correlation

If $\hat{\rho}$ is significantly different from zero, we reject the null and conclude that the error terms exhibit serial correlation.

7.2 Fixed Effects Estimation

Fixed Effects Transformation

Consider the following panel data model with a single expl variable and an unobserved fixed effect a_i:

$$y_{it} = \beta_1 x_{it} + a_i + u_{it}, \quad t = 1, 2, \ldots, T$$

We can eliminate a_i by *time-demeaning* each variable. First, take the time average for each unit i:

$$\bar{y}_i = \beta_1 \bar{x}_i + a_i + \bar{u}_i$$

Then subtract this average from the original equation to obtain:

$$y_{it} - \bar{y}_i = \beta_1(x_{it} - \bar{x}_i) + (u_{it} - \bar{u}_i)$$

Or, using notation for time-demeaned variables:

$$\ddot{y}_{it} = \beta_1 \ddot{x}_{it} + \ddot{u}_{it}$$

This is the **Fixed Effects (FE) estimator**, also known as the *within estimator*, as it exploits variation *within* each cross-sectional unit over time.

Extensions and Other Estimators

Adding more regressors causes no major changes. For instance, with multiple explanatory variables:

$$y_{it} = \beta_1 x_{it1} + \beta_2 x_{it2} + \cdots + \beta_k x_{itk} + a_i + u_{it}$$

We time-demean each regressor and estimate the model via pooled OLS on the transformed variables:

$$\ddot{y}_{it} = \beta_1 \ddot{x}_{it1} + \beta_2 \ddot{x}_{it2} + \cdots + \beta_k \ddot{x}_{itk} + \ddot{u}_{it}$$

The FE transformation removes any time-invariant explanatory variables since they disappear during the time-demeaning step.

An alternative is the **Between Estimator (BE)**, which regresses time-averaged outcomes on time-averaged regressors:

$$\bar{y}_i = \beta_1 \bar{x}_i + a_i + \bar{u}_i$$

However, the BE estimator is biased if a_i is correlated with $\bar{x}_i$, which is usually the case. Therefore, BE is generally not recommended unless strict exogeneity holds.

Assumptions and Properties

Under the assumption that the explanatory variables are strictly exogenous:

$$\text{Cov}(x_{it}, u_{is}) = 0 \quad \text{for all } s, t$$

the FE estimator is unbiased and consistent. This also implies that the error term u_{it} should be:

- Homoskedastic
- Serially uncorrelated across time

The FE transformation, like first differencing, allows arbitrary correlation between a_i and the regressors. However, any variable that does not vary over time (e.g. gender, country) is swept away and cannot be estimated.

FE vs. FD: When to Use Each

- For $T = 2$, FE and FD yield identical results.
- For $T \geq 3$:
  - Use **FE** if u_{it} is serially uncorrelated — more efficient.
  - Use **FD** if u_{it} is serially correlated — differencing reduces serial correlation.
- **Unbalanced panels**: FE is valid if missing data is unrelated to u_{it}; otherwise, bias may occur (e.g., attrition).
- **Large T, small N**: FE becomes sensitive to non-normality, heteroskedasticity, and serial correlation — interpret results with caution.

7.3 Random Effects Estimation

We consider the standard unobserved effects model:

$$y_{it} = \beta_0 + \beta_1 x_{it1} + \ldots + \beta_k x_{itk} + a_i + u_{it}$$

with a_i as an unobserved time-invariant effect.

The **random effects (RE)** model assumes that a_i is uncorrelated with all explanatory variables in all time periods:

$$\text{Cov}(x_{itj}, a_i) = 0 \quad \text{for all } t = 1, \ldots, T; \; j = 1, \ldots, k$$

Under this assumption, we define the composite error:

$$v_{it} = a_i + u_{it}$$

and rewrite the model as:

$$y_{it} = \beta_0 + \beta_1 x_{it1} + \ldots + \beta_k x_{itk} + v_{it}$$

Implication: Serial Correlation in Errors Because a_i is constant over time and shared across all t, the composite error v_{it} is serially correlated:

$$\text{Corr}(v_{it}, v_{is}) = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_u^2} > 0 \quad \text{for } t \neq s$$

$$\sigma_v^2 = \sigma_a^2 + \sigma_u^2$$

7.3.1 GLS Estimation via Quasi-Demeaning

To eliminate serial correlation in v_{it}, we apply a Generalized Least Squares (GLS) transformation. Define:

$$\theta = 1 - \left(\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_a^2} \right)^{1/2}, \textit{where } \theta \in (0,1)$$

Then the quasi-demeaned model becomes:

$$y_{it} - \theta \bar{y}_i = \beta_0(1-\theta) + \beta_1(x_{it1} - \theta \bar{x}_{i1}) + \ldots + \beta_k(x_{itk} - \theta \bar{x}_{ik}) + (v_{it} - \theta \bar{v}_i)$$

which denote time-averages.

The Generalized Least Squares (GLS) estimator is simply the pooled OLS estimator applied to this quasi-demeaned model. This allows us to correct for serial correlation in the composite error $v_{it} = a_i + u_{it}$, which arises when a_i is present in each period and hence induces correlation across time. Quasi-demeaning is crucial for obtaining efficient estimates under the RE assumptions.

7.3.2 Estimating Variance Components

We estimate σ_a^2 from pooled OLS residuals $\hat{v}_{it}$ via:

$$\hat{\sigma}_a^2 = \left[\frac{NT(T-1)}{2} - (k+1) \right]^{-1} \sum_{i=1}^{N} \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} \hat{v}_{it}\hat{v}_{is}$$

Then estimate:

$$\hat{\sigma}_u^2 = \hat{\sigma}_v^2 - \hat{\sigma}_a^2$$

with $\hat{\sigma}_v^2$ as the residual variance from pooled OLS.

Properties of the RE Estimator

The Random Effects (RE) estimator assumes that the unobserved effect a_i is uncorrelated with the regressors across all time periods. Under this assumption, RE is consistent (though not unbiased) and asymptotically normally distributed as $N \to \infty$ with fixed T. While it has been used when T is large and N is small, its statistical properties in that setting remain uncertain.

RE vs. FE Estimation

The RE estimator uses a quasi-demeaned version of the model that lies between pooled OLS and FE. Specifically:

- When $\theta = 0$, RE becomes equivalent to pooled OLS.
- When $\theta = 1$, RE coincides with FE.

In practice, $\hat{\theta}$ lies strictly between 0 and 1. When the variance of a_i is small relative to the variance of u_{it}, RE estimates are close to those from pooled OLS. Conversely, if the variance of a_i dominates, RE approximates FE. As T increases, $\hat{\theta} \to 1$, and the two estimators become increasingly similar.

Choosing Between RE and FE

FE allows for arbitrary correlation between a_i and the regressors x_{it}, making it more reliable for causal inference under minimal assumptions. In contrast, RE is appropriate in special cases—e.g., when the key explanatory variable is constant over time or set experimentally and plausibly uncorrelated with a_i. A typical example is random assignment in education policy studies.

Hausman Test: To decide formally between RE and FE, one can use the Hausman test to check whether the difference in coefficients is statistically significant. If so, the RE assumption of no correlation between x_{it} and a_i is violated, and FE is preferred.

Chapter 8

Lecture 9

8.1 Difference-in-Differences (DiD)

Difference-in-differences (DiD) is a quasi-experimental design used to estimate causal effects using longitudinal data. It constructs a counterfactual by comparing changes in outcomes over time between a treated group and a control group.

DiD is particularly useful for evaluating the impact of interventions or policies—such as a change in minimum wage or public health regulations—when randomized experiments are not feasible.

Application: Minimum Wage and Employment

Card and Krueger (1994) applied DiD to assess the effect of a minimum wage increase in New Jersey (NJ). On April 1, 1992, NJ raised its minimum wage from \$4.25 to \$5.05. To evaluate the employment impact, they collected data from fast food restaurants in NJ and a neighboring state, Pennsylvania (PA), where the minimum wage remained constant.

They observed employment levels in February (before the change) and in November (after the change), providing a natural pre-post treatment window. Pennsylvania served as the control group.

Let:

$$
\begin{aligned}
Y_{1it} &= \text{employment at restaurant } i \text{ in period } t \text{ under high minimum wage,} \\
Y_{0it} &= \text{employment at restaurant } i \text{ in period } t \text{ under low minimum wage.}
\end{aligned}
$$

These are *potential outcomes*. We only observe one per unit and time. For example, in NJ in November 1992, we observe Y_{1it}, while in PA, we observe Y_{0it}.

Assume the following structure for untreated outcomes:

$$\mathbb{E}[Y_{0ist} \mid s, t] = \gamma_s + \lambda_t$$

where $s \in \{\text{PA}, \text{NJ}\}$ indexes states and $t \in \{\text{Feb}, \text{Nov}\}$ indexes time. This model decomposes the untreated outcome into an additive state effect and time effect, just as in a fixed effects regression.

Let D_{st} be a dummy equal to 1 if state s at time t is subject to the treatment (i.e., NJ in November 1992), and 0 otherwise. Assume:

$$\mathbb{E}[Y_{1ist} - Y_{0ist} \mid s, t] = \beta$$

Then the regression model becomes:

$$Y_{ist} = \gamma_s + \lambda_t + \beta D_{st} + \varepsilon_{ist}, \quad \mathbb{E}[\varepsilon_{ist} \mid s, t] = 0$$

The causal effect β is identified via:

$$\begin{aligned}&(\mathbb{E}[Y_{ist} \mid s = \text{NJ}, t = \text{Nov}] - \mathbb{E}[Y_{ist} \mid s = \text{NJ}, t = \text{Feb}]) \\ &\quad - (\mathbb{E}[Y_{ist} \mid s = \text{PA}, t = \text{Nov}] - \mathbb{E}[Y_{ist} \mid s = \text{PA}, t = \text{Feb}]) = \beta\end{aligned}$$

Empirical Evidence

The table below shows the average full-time-equivalent (FTE) employment per store:

	PA	NJ	Difference (NJ - PA)
Before (Feb 1992)	23.33	20.44	-2.89
After (Nov 1992)	21.17	21.03	-0.14
Change	-2.16	+0.59	**+2.76**

DiD estimate +2.76: employment in NJ improved, while it dropped in PA. The resulting positive DiD is surprising if one expects higher min wages to reduce employment due to standard labor demand theory.

8.1.1 Parallel Trends Assumption in DiD

A key assumption of the DiD design is the **parallel trends assumption**. It states that, in the absence of treatment, the outcome for the treated and control groups would have evolved similarly over time.

Challenge: NJ vs. PA While the original Card and Krueger study found only a small employment decline in Pennsylvania (PA) and stable employment in New Jersey (NJ) between February and November 1992, further administrative data revealed larger year-to-year fluctuations.

In particular, PA employment dropped relative to NJ even before the 1992 policy change, especially between 1991 and 1993. This casts doubt on the validity of PA as a counterfactual for NJ since both states did not appear to follow parallel pre-trends.

Key Takeaway:

Evidence over multiple time periods can support/invalidate the parallel trends assumption.
The credibility of a DiD estimate hinges on the plausibility of parallel trends.
Pre-treatment data are essential for testing this assumption and validating the control group as a valid counterfactual.

8.2 DiD: Regression Formulation

A regression-based framework allows us to estimate treatment effects, compute standard errors, and include controls or more complex setups.

Let NJ_s be a dummy for New Jersey, and d_t a time dummy equal to 1 in November (i.e., after the minimum wage change). Then the DiD regression is:

$$Y_{ist} = \alpha + \gamma NJ_s + \lambda d_t + \beta(NJ_s \cdot d_t) + \varepsilon_{ist}$$

- α: Baseline employment in PA before the policy
- γ: Difference between NJ and PA before the policy
- λ: Time effect (Nov vs. Feb) common to both states
- β: The DiD estimate, i.e., the causal effect of the policy

This saturated model has one term for each of the four group-time combinations.

We can map the regression parameters to mean differences:

$$\begin{aligned}
\alpha &= \mathbb{E}[Y_{ist} \mid s = \text{PA}, t = \text{Feb}] = \gamma_{\text{PA}} + \lambda_{\text{Feb}} \\
\gamma &= \mathbb{E}[Y_{ist} \mid s = \text{NJ}, t = \text{Feb}] - \mathbb{E}[Y_{ist} \mid s = \text{PA}, t = \text{Feb}] \\
\lambda &= \mathbb{E}[Y_{ist} \mid s = \text{PA}, t = \text{Nov}] - \mathbb{E}[Y_{ist} \mid s = \text{PA}, t = \text{Feb}] \\
\beta &= \{\mathbb{E}[Y_{ist} \mid s = \text{NJ}, t = \text{Nov}] - \mathbb{E}[Y_{ist} \mid s = \text{NJ}, t = \text{Feb}]\} \\
&\quad - \{\mathbb{E}[Y_{ist} \mid s = \text{PA}, t = \text{Nov}] - \mathbb{E}[Y_{ist} \mid s = \text{PA}, t = \text{Feb}]\}
\end{aligned}$$

It allows is to include controls or more complex setups like an extension to multiple states or periods, or inclusion of additional covariates.

For example, we can write the counterfactual outcome as:

$$\mathbb{E}[Y_{0ist} \mid s, t, X_{st}] = \gamma_s + \lambda_t + X'_{st}\delta$$

where X_{st} is a vector of state-and-time-varying covariates.

With multiple years, we can check the identifying assumption (no anticipation) by including *leads and lags* of the treatment:

$$Y_{ist} = \gamma_s + \lambda_t + \sum_{\tau=0}^{m} \beta_{-\tau} D_{s,t-\tau} + \sum_{\tau=1}^{q} \beta_{+\tau} D_{s,t+\tau} + X'_{ist}\delta + \varepsilon_{ist}$$

If leads $\beta_{+\tau}$ are insignificant, this supports the parallel trends assumption.

Control Group Selection:

If treatment affects group composition, results may be misleading, that's why you have to choose it carefully. For example, in evaluating the effect of generous welfare benefits on labor supply, one must consider that individuals with weak labor market attachment may migrate to states with more generous assistance. This migration, induced by the policy, can bias the DiD estimate